INCH-POUND

MIL-T-55255A(CR)
AMENDMENT 1
17 February 1989

MILITARY SPECIFICATION

TERMINAL, TELEGRAPH-TELEPHONE AN/TCC-29( ) AND UNITS OF

This amendment forms a part of MIL-T-55255A, dated 9 January 1973, and is approved for use by the Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of defense.

PAGE 1

Add paragraphs 2.1 & 2.1.1:

' 2.1. Government documents.'

' 2.1.1. Specifications and standards. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.'

Under ' Federal Specifications ', add the followings:

' L-P-378 Plastic Sheet and Strip, Thin Gage, Polyplefin'
' L-T-871 Twine, Fibrous Cotton, Wrapping'

PAGE 2

Under ' Military Specifications', add the followings:

' MIL-P-19644 Plastic Molding Material (Polystyrene Foam, Expanded Bead)'
' MIL-P-26514 Polyurethane Foam, Rigid or Flexible, for Packing'

Add ' FEDERAL STANDARD, FED-STD-101 Preservation, Packaging, and Packing Materials, Test Procedures'

Add paragraph 2.1.2:

' 2.1.2. Other Government documents, drawings and publications. - The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be those in effect on the date of the solicitation.'

Under the ' DRAWINGS ', add this drawing number,' SPI-1G00025 Special Packaging Instructions.'

Add paragraph 2.2:

' 2.2. Order of precedence. - In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specification, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.'

AMSC N/A FSC 5805

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3.8.3: Delete and substitute:

' 3.8.3. Vf output level. - The vf output level of the terminal into a 600 ohm resistive load shall be -10dbm + 4db -2db. The difference in level between mark and space frequencies shall not exceed 1db (See 4.10.3)'.

3.8.4: Delete and substitute:

' 3.8.4. Distortion. - The total peak distortion introduced into a teletypewriter signal being processed by the terminal shall not exceed five percent. This requirement applies at the teletypewriter speeds of 37.5, 50, 75 and 150 baud. No internal adjustment of the terminal will be allowed when switching between modes (MODE ONE provides for operation at 37.5, 50 and 75 baud and MODE TWO provides for operation at 150 baud).'

3.8.7: Delete and substitute:

' 3.8.7. Send circuit impedance. - The output impedance of the terminal measured across the 2W 4WR binding posts and across the 4WS TEL binding posts shall be 600 + 35% -20% ohms when transmitting a 1232.5 +/- 3 Hz signal (See 4.10.7).'

3.8.12: Delete and substitute:

' 3.8.12. Receive circuit impedance. - The impedance measured across the 2W 4WR binding posts, with power applied to the terminal, shall be 600 + 35% -20% ohms over the frequency range 1000 to 2000 Hz (See 4.10.12).'

3.8.13: Delete and substitute:

' 3.8.13. Threshold circuit. - The receive circuit output shall switch to steady mark signal, and the THRESHOLD lamp shall light when the input signal level falls to -25dbm +/- 5db with the high-low sensitivity strap attached in the low sensitivity position or -50dbm +/- 5db with the strap attached in the high sensitivity position. The output shall reach steady mark condition not more than 2 milliseconds after the input level falls to the threshold level (to avoid activating teletypewriters connected to the receive circuit that are in motor stop condition) (See 4.10.13).'

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3.8.14.1.1: Delete and substitute:

" 3.8.14.1.1. Break-in signal level. - The break-in signal level shall be -10dbm + 4db -2db with the BOOST switch in the lower position and 0dbm + 4db -2db with the BOOST switch in the upper (operated) position (See 4.10.14)."

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3.8.16.2: Delete and substitute:

" 3.8.16.2. Vf ring signal output. - The terminal shall transmit a 1232.5 Hz (space), -10dbm + 4db - 2db signal into a 600 ohm resistive load when the RING switch is operated with the VF 20 switch in the VF position (See 4.10.22)."

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3.10.3: Delete and substitute:

" 3.10.3. Vf output level. - The vf output level of the converter into a 600 ohm resistive load shall be -10dbm +/- 2db when triggered by a 20 Hz, 90 volt (180 volt peak-to-peak) input (See 4.12.2)."

3.10.5: Delete and substitute:

" 3.10.5. Vf sensitivity. - The vf signal input level required to trigger the converter 20 Hz output shall be as shown by Table II (See 4.12.3)."

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Add paragraph 3.10.13:

" 3.10.13. Message transmission. - The CV-425/U Converters shall not generate a ring signal (20Hz) and the TH-22/TG Terminal's audible alarm shall not sound while transmitting a repeated blank character in baudot (5-level) code at 45 and 75 baud in (TH-22) Mode 1, and 150 baud in (TH-22) Mode 2."

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Delete paragraph 3.11.2 in its entirety.

PAGE 13

Delete paragraphs 3.11.5.1 & 3.11.5.2 in their entirety.

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3.2: Delete and substitute:

" 3.2. SYSTEM SAFETY REQUIREMENTS."

Add paragraphs 3.2.1 & 3.2.2:

" 3.2.1. Personnel Hazards. - Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454, Requirement 1, shall be made applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly process, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with this requirement has been achieved shall be through a visual inspection (See 4.8.1)."

" 3.2.2. Edge Rounding Exposed. - Exposed edges and corners shall be rounded sufficiently to minimize laceration/puncture hazards, the design goal being a minimum radius of 0.04 inch (1 mm) for edges and a minimum of 0.5 inch (12.7 mm) for corners."

4.2(a): Delete and substitute:

" (a) First article inspection. (See 4.3) Does not include packaging."

4.2(c)(1): Delete and substitute:

" (1) Quality conformance inspection of equipment before packaging. (See 4.5)."

Delete paragraph 4.2(c)(2) in its entirety.

Add paragraph 4.2(d):

" 4.2(d) Packaging inspection (See 4.18)."

Table III, line 6, Group C inspection: Delete " Vibration, Humidity, Bounce, loose cargo."

Table III, line 6, Group C inspection: Delete " Reqt Para. 3.11.5.1, 3.11.2, and 3.11.5.2."

Table III, line 6, Group C inspection: Delete " Test Para. 4.17, 4.14, and 4.18."

4.5: Delete "preparation for delivery" and substitute "packaging".

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Delete paragraphs 4.7, 4.7.1 thru 4.7.4 in their entirety.

Add paragraph 4.8.1:

* 4.8.1. Visual and Mechanical Inspection. - An inspection shall be performed to verify compliance with those portions of 3.2 (Systems Safety Engineering) which can be determined visually.*

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4.10.14: Delete and substitute:

* 4.10.14 Break-in transmission signal level and lock out tests. - Connect the terminal as shown by Figure 12 but with the attenuator switched out the circuit. Set the message generator to transmit undistorted test message. Operate the RESET switches of both terminals and note that neither audible alarm is operating. Operate the BREAK IN switch of the unit under test, and note that the audible alarm of the unit not under test sounds. Note that until the BREAK IN switch is released, the teletypewriter runs closed. Measure the output frequency and level of the unit under test with the BREAK IN switch operated with and without the BOOST switch operated to determine conformance to 3.8.14.1 and 3.8.14.1.1. Release the BREAK IN switch; note that both audible alarms sound and that the teletypewriter is in open (steady space) condition (See 3.8.14.1.2). Operate the RESET switch of the unit not under test; note that both audible alarms cease to sound and that the teletypewriter resumes printing the test message. Reverse the unit under test from the receiving to the transmitting position in Figure 12. Operate the BREAK IN switch of the unit not under test with the break-in disable strap installed in the unit under test to determine whether detection of the break-in signal has been prevented in accordance with 3.8.14.1.*

4.10.16: Delete and substitute:

* 4.10.16 Break-in reception and sensitivity test. - Connect the terminal as shown by Figure 14 but with the attenuator switched out of the system. Set the message generator to transmit undistorted test message at 75 baud. Operate the RESET of both terminals, and note that neither audible alarm is operating. Operate the BREAK IN switch of the unit not under test, and note that the audible alarm of the unit under test sounds and that a space (open) signal is present on the REC jack of the unit under test. Release the BREAK IN switch and measure the vf output frequency and level of the unit under test to determine conformance to 3.8.14.3. Repeat the test with the attenuator set to provide a -30dbm and a -32dbm signal to the unit under test to determine conformance to 3.8.14.4.*

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Delete paragraphs 4.14 and 4.17 in their entirety.

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4.18: Delete and substitute:

" 4.18. Packaging inspection. - Packaging inspection requirements specified herein are classified as follows:"

Add paragraphs 4.18a, 4.18b, 4.18.1, 4.18.2, 4.18.2.1 thru 4.18.2.3.

" a. First Article Inspection of Packaging."

" b. Quality Conformance Inspection of Packaging."

" 4.18.1 First Article Inspection of Packaging. - Unless otherwise specified in the contract, First Article Inspection of Packaging shall be in accordance with the Unit Pack Design Validation Requirements of MIL-P-116."

" 4.18.2 Quality Conformance Inspection of Packaging."

" 4.18.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification."

" 4.18.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with Group A and B Quality Conformance Inspection Requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein."

" 4.18.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in Table VIII, "PACKING INSPECTION PROVISIONS." Lot formation shall consist of all packs made of the same materials during identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and Acceptable Quality level of 4.0 percent defective."

Add Table VIII.

TABLE VIII. PACKING INSPECTION PROVISIONS.

NO.	CHARACTERISTIC	METHOD OF INSPECTION
101	Intermediate container not as specified	Visual
102	Improper closure of intermediate container	Visual
103	Shipping containers not in accordance with specification	Visual
104	Excessive tube	Visual
105	Improper blocking and bracing	Visual
106	Closure not in accordance with specification	Visual
107	Weight and size exceed container limitations	Weight & Measure
108	Strapping not in accordance with specification, incorrectly applied, omitted	Visual
109	Marking omitted, incorrect, or illegible	Visual

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6.2(c) & (e): Delete and substitute:

" (c) Level A or B preservation and packing (See Section 5)."

" (e) When performing First Article Inspection rough handling tests are not required."

Add paragraph 6.2(j):

" (j) First Article Packaging Inspection test reports require Acquisition Activity approval prior to production unit packing."

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Add paragraphs 6.7 & 6.8;

" 6.7 Level B preservation. - When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage, and handlings."

" 6.8 Environmental. - Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods."

PAGE 35

Figure 1, line 5: Delete and substitute:

" Distortion Analyzer - Data Products of New England Model DMS-303A."

Figure 1, line 15: Delete and substitute:

" Test Message Generator - Data Products of New England Model PG-303A."

PAGE 36

Delete and substitute Figure 2. (See attached page)

PAGE 37

Figure 3, add Note 3:

" 3. Set attenuator at 3 db."

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Figure 6, Note 2: Delete and substitute:

" Set MODE ONE/MODE TWO Switch to MODE TWO for operation at 150 baud and MODE ONE at 37.5, 50 and 75 baud."

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Figure 7, add Notes 5 & 6:

" 5. For the signal delay test, change the TH-22 4W/2W/TEL switch position from 4W to 2W."

" 6. For the signal delay test, the message generator output shall be alternate mark space at 75 baud."

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Figure 9, add Note 2:

" 2. Place an open plug into the TH-22 SEND jack for transmission of a space signal."

PAGE 45

Figure 11, Delete and substitute:

" 4W 2W TEL- 4W."

Figure 11, Note 4, Delete and substitute:

" 4. Attenuator required for Receive Circuit Sensitivity Test only. To be adjusted in accordance with test paragraph. Operate BOOST switch to obtain 0dbm signal level."

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Figure 12, add Note 3:

" 3. If F-316( )/U Filter is not part of the contract, connect the 2W/4WR terminals of the TH-22 directly to the attenuator."

PAGE 67

Add Figure 33.

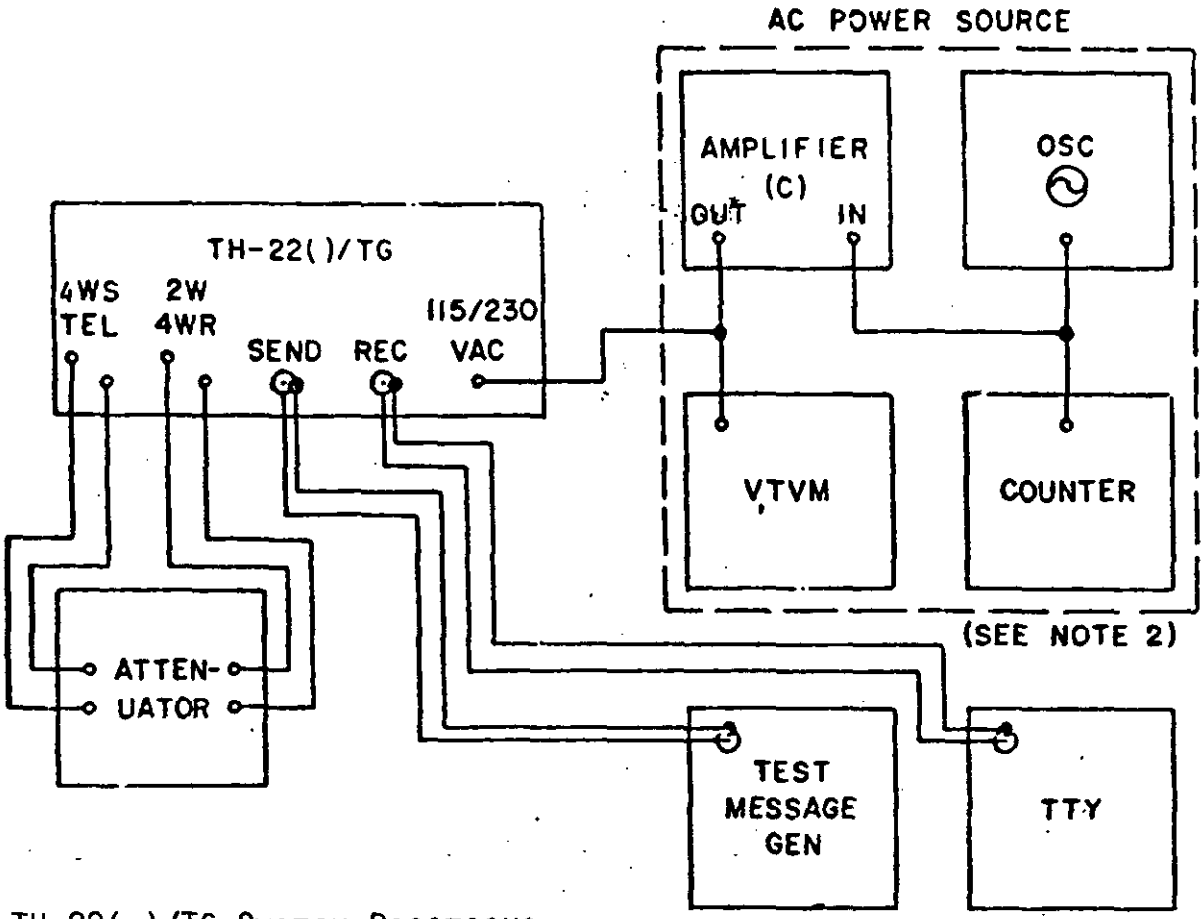
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Add Figure 34.

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TH-22( )/TG Switch Positions

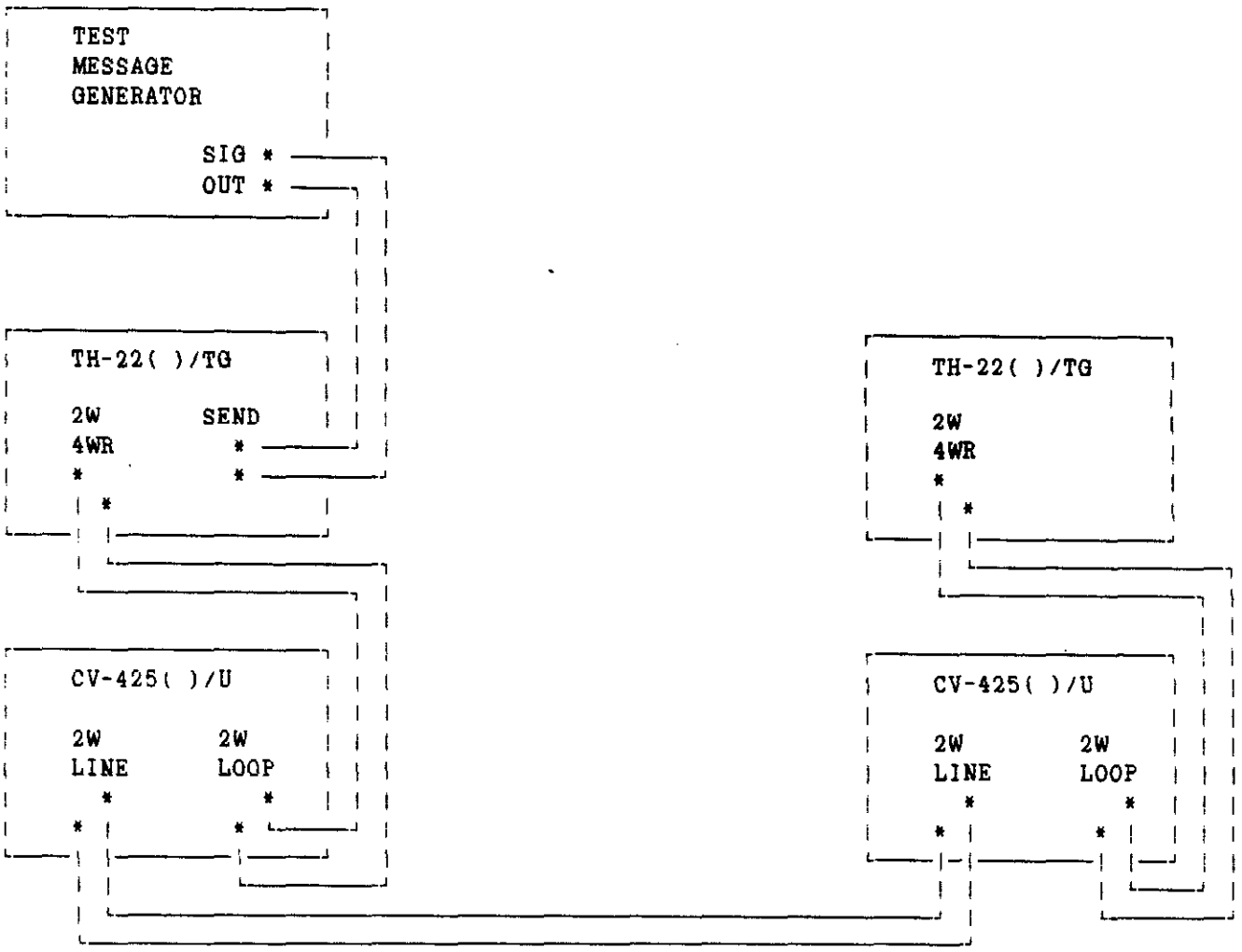
115VAC/230VAC/26VDC - 115VAC & 230VAC
ON OFF - ON
NORM REG SEND - NORM
4w 2w TEL - 4w
VF 20 - Either
MODE ONE MODE TWO - MODE ONE

Notes

1. See Figure 1 for test equipment description.
2. Power source oscillator frequency and amplifier output level to be adjusted in accordance with power source test paragraph.
3. Set attenuator at 3 db.

Figure 2 TH-22( )/TG
AC Power Source Test

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Note 1: A2 card terminals 5 & 6 strapped for generator drive keyed input, A2 card terminals 5 & 6 open for generator 20 ma neutral keying.

Equipment Control Settings

Test Message Generator	TH-22( )/TG	CV-425( )/TG
DISTORTION: None	NORM-REC-SEND: NORM	2W-4W: 2W
BAUD RATE: 45, 75, 150 (See 3.10.13)	4W-2W-TEL: 2W	TP-TG: TG
PATTERN SELECTOR: SELECTED PULSES	MODE ONE-MODE TWO: (See 3.10.13)	SENSITIVITY: ANY
SELECTED CHARACTER BITS 1 thru 5: SPACE	VF-20: 20	POWER: ON
	POWER: ON	

Figure 33. CV-425 ( ) /U Message Transmission Test

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